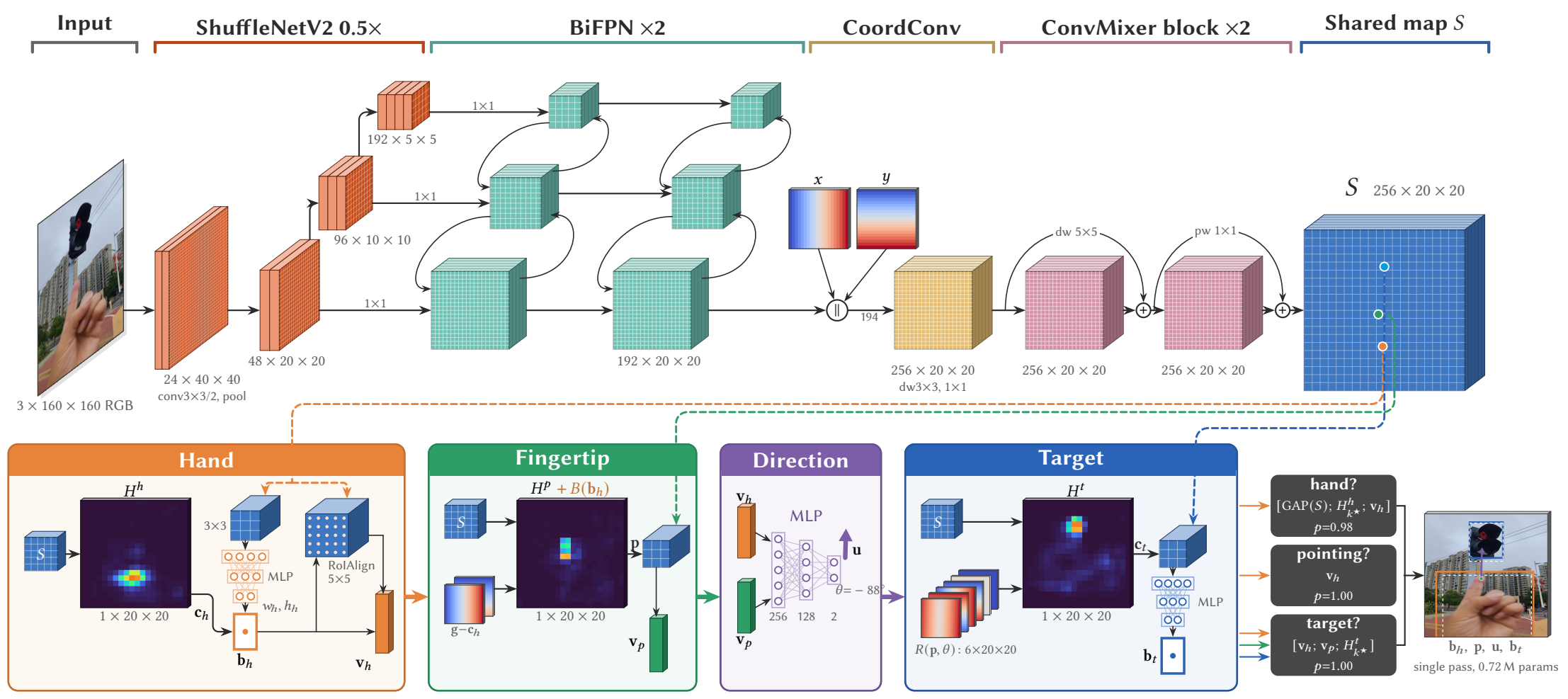

Input
ShuffleNetV2 0.5×
BiFPN ×2
CoordConv
ConvMixer block ×2
Shared map S
3 × 160 × 160 RGB
24 × 40 × 40
conv3×3/2, pool
48 × 20 × 20
96 × 10 × 10
192 × 5 × 5
1×1
192 × 20 × 20
x
y
194
256 × 20 × 20
dw3×3, 1×1
dw 5×5
256 × 20 × 20
pw 1×1
256 × 20 × 20
S 256 × 20 × 20
Hand
Fingertip
Direction
Target
MLP
RoIAlign 5×5
3×3
1 × 20 × 20
R(p, θ) : 6×20×20
256 128 2
hand?
[GAP(S); H^h_k★; v_h]
p=0.98
pointing?
v_h
p=1.00
target?
[v_h; v_p; H^t_k★]
p=1.00
b_h, p, u, b_t
single pass, 0.72 M params